Discrete Mathematics Spring 2026

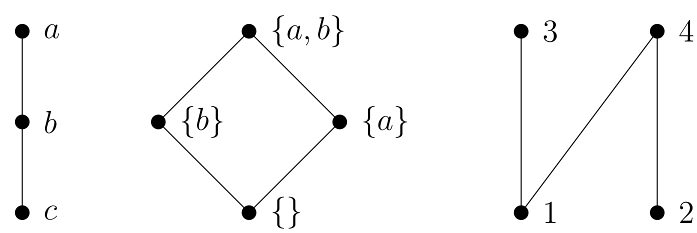
Homework 7: Posets

Due Sunday March 29 @11:59:00pm Eastern

Show all your work to receive full credit

1. **Reading Hasse diagrams**

   Consider the three following Hasse diagrams representing three posets:



(A) (B) (C)

   (a) For each poset, list all the ordered pairs that belong to the relation. Recall that Hasse diagrams omit self-loops (reflexivity) and transitive edges — you must include these when listing all pairs.

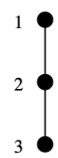
   (b) For each poset, identify all minimal elements, maximal elements, the least element (if it exists), and the greatest element (if it exists). Justify each answer.

   (c) For poset (C), compute the GLB and LUB of each pair below. If a GLB or LUB does not exist, say so and explain why.

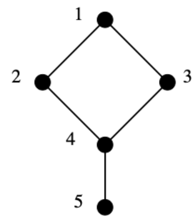
      1. The GLB and LUB of 1 and 2.
      2. The GLB and LUB of 3 and 4.
      3. The GLB and LUB of 1 and 4.

2. **Identifying lattices**

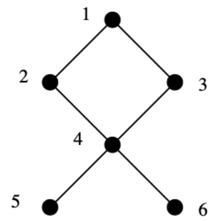
   Which of the following posets are lattices? For each non-lattice, identify a specific pair of elements with no GLB or no LUB and explain why it does not exist. No justification is required for the lattices.



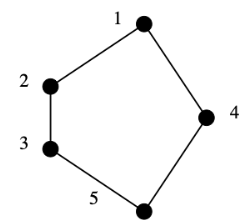
(A)



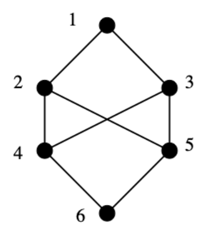
(B)



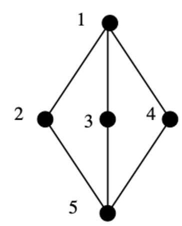
(C)



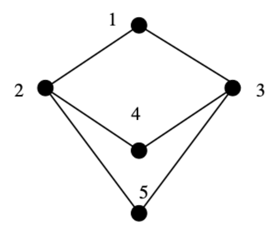
(D)



(E)



(F)



(G)

3. **Computing in lattices**

   You may assume both posets below are valid posets (reflexivity, antisymmetry, and transitivity hold in each case).

   (a) Let $(S, \preccurlyeq)$ be a poset such that:

   $$S = \{1, 2, 3, 5, 6, 10, 15, 30\}, \qquad x \preccurlyeq y \text{ IFF } x \mid y$$

   i. Draw the Hasse diagram of $(S, \preccurlyeq)$.
   ii. Is this poset a lattice? Justify your answer.
   iii. If this poset is a lattice, what do the operations meet $\wedge$ and join $\vee$ represent in this poset? Be specific and provide at least two examples.

   (b) Let $(S, \preccurlyeq)$ be a poset such that:

   $$S = P(\{1, 2, 3\}) \text{ (powerset of } \{1, 2, 3\}), \qquad x \preccurlyeq y \text{ IFF } x \subseteq y$$

   i. Draw the Hasse diagram of $(S, \preccurlyeq)$.
   ii. Is this poset a lattice? Justify your answer.
   iii. If this poset is a lattice, what do the operations meet $\wedge$ and join $\vee$ represent in this poset? Be specific and provide at least two examples.